



Networking Concept

Private IP, Public IP, Bastion Host & Connection Flow

◆ 1. What is IP Address?

👉 IP address is a **unique number** used to identify a device on a network.

Example:

- 192.168.54.1
- 23.45.223.5

👉 Just like:

- Home address = where you live
 - IP address = where your computer lives
-

◆ 2. Types of IP Address



Public IP

👉 Example: 23.45.223.5

- Works on internet
 - Accessible from anywhere
 - Used by websites, servers
-



Private IP

👉 Example: 192.168.54.1

- Used inside local network / VPC
 - Not accessible from internet directly ❌
 - Used for internal communication
-

◆ 3. Can Private IP and Public IP Connect?

❌ Direct Connection (Not Possible)

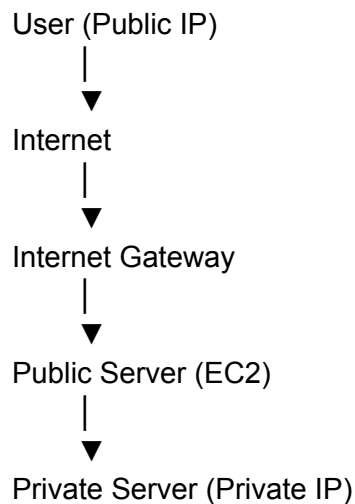
23.45.223.5 ❌ → 192.168.54.1

👉 Reason:

- Private IP is hidden
 - Internet cannot see it
-

✅ Indirect Connection (Possible)

👉 Using a middle server



◆ 4. What is Bastion Host?

👉 **Definition:**

A Bastion Host is a **public server** used to access **private servers** securely.

◆ **Simple Meaning**

👉 Bastion Host = **middle server (gateway)**

◆ **Flow**

User → Internet → Bastion Host → Private Server

◆ **Real-Life Example**

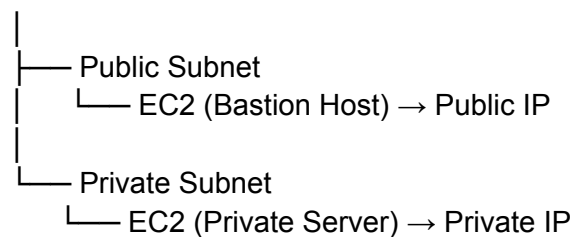
- Private Server = your room 🚪
- Bastion Host = main gate 🚪
- User = outsider 🌐

👉 Outsider:

- Cannot enter room directly ❌
 - Must go through gate ✅
-

◆ **5. AWS Architecture Example**

VPC



◆ 6. How Connection Works (Step-by-Step)

👉 Step 1

User sends request (from internet)

👉 Step 2

Request enters VPC through Internet Gateway

👉 Step 3

Request reaches Public EC2 (Bastion Host)

👉 Step 4

Bastion Host forwards request to Private Server

👉 Step 5

Private Server sends response back

◆ 7. Commands Example (SSH)

Step 1: Connect to Bastion Host

```
ssh ec2-user@public-ip
```

Step 2: Connect to Private Server

```
ssh ec2-user@private-ip
```

◆ 8. Important Networking Components



Route Table

- Decides where traffic goes



NACL (Network ACL)

- Subnet-level firewall
- Allow/Deny traffic



Security Group

- Instance-level firewall
-

◆ 9. Private IP to Private IP Connection



Same Network

192.168.23.1 ⇌ 172.20.43.243

✓ Works if:

- Same VPC
 - Routing exists
 - Security allows
-



Different Network

👉 No direct connection



With Setup

- VPN
- VPC Peering

- Routing
-

◆ 10. Important Rule



"Not all IPs can connect. Connection depends on network, routing, and permissions."

◆ 11. Key Points to Remember

- Private IP → no direct internet access
 - Public IP → internet accessible
 - Bastion Host → secure access to private servers
 - Route Table → decides path
 - NACL → subnet firewall
 - Security Group → instance firewall
-



Final Summary



- Direct connection (Public → Private) = ❌
 - Using Bastion Host = ✅
 - Private to Private = depends on network
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One Line (Interview)



"A Bastion Host is a public server used to securely access private instances in a VPC."
